

Cryptocurrency Price Prediction Project Report

1. Project Overview

The objective of this project is to **predict the next-day price of cryptocurrencies (BTC and ETH)** using historical data and machine learning models.

Given the volatile nature of crypto markets, the project aims to analyze patterns, evaluate model performance, and perform inference on **unseen 2026 data**.

2. Data Collection

- **Data Source:** Binance Futures API (`fapi.binance.com`)
- **Assets:** BTCUSDT and ETHUSDT
- **Time Period:** Daily candles from 2024-01-01 to 2025-12-31
- **Data Points:** 757 records per asset
- **Inference Data:** Last 27 records (2026) separated for evaluation

Columns Extracted:

`open_time, open, high, low, close, volume, close_time, quote_asset_volume, num_trades, taker_buy_base_volume, taker_buy_quote_volume`

Special Note:

- For inference, the last 27 records were isolated to simulate **unseen future data**.
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3. Data Preprocessing

Feature Selection & Engineering

For price prediction, the following features were created:

- `open, volume` – raw data features
- `return_1d` – daily return: `(close - open) / open`
- `vol_3d` – rolling 3-day volatility: `(high - low).rolling(3).mean()`
- `ma_3d` & `ma_7d` – 3-day and 7-day moving averages of `close`
- Missing values were **forward-filled**, and the dataset was **sorted chronologically**

Scaling & Transformation

- **StandardScaler** was used for feature scaling
- Log transformation was applied safely to **positive-only features**

Reasoning: This ensures consistency with training data and avoids data leakage.

4. Model Selection & Training

Three regression models were trained for each asset:

1. **Linear Regression (LR)** – baseline model
2. **Decision Tree Regressor (DT)** – handles non-linear relationships
3. **Random Forest Regressor (RF)** – ensemble model, generally robust to volatility

Train/Test Split: 80% training, 20% testing (chronological)

BTC Evaluation Metrics (2024–2025)

Model	MAE	MSE	RMSE	R ²
Linear Regression	3,347.68	15,819,312.04	3,977.35	0.8865
Decision Tree	3,079.67	14,168,131.89	3,764.06	0.8984
Random Forest	2,273.99	8,074,862.46	2,841.63	0.9421

ETH Evaluation Metrics (2024–2025)

Model	MAE	MSE	RMSE	R ²
Linear Regression	203.81	65,416.73	255.77	0.8360
Decision Tree	288.69	142,948.93	378.09	0.6417
Random Forest	298.99	159,242.73	399.05	0.6008

Model Selection:

- BTC: Random Forest (highest R², lowest RMSE)
- ETH: Linear Regression (highest R²)

Models and corresponding scalers were saved as `.pkl` files for inference.

5. Inference on 2026 Data

Data: Last 27 records separated during data collection.

BTC Predictions

Metric	Value
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MAE	1,751.23
MSE	5,729,009.56
RMSE	2,393.54
R ²	0.2419

ETH Predictions

Metric	Value
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MAE	71.77
MSE	9,470.02
RMSE	97.31
R ²	0.6204

Interpretation:

- BTC predictions show **low R² (0.24)** due to **high volatility and non-stationary nature** of the market.
- ETH predictions are moderately better (R² ~0.62), reflecting slightly more predictable price movements.
- Despite high evaluation metrics during training, **crypto price prediction remains challenging**, and inference results must be interpreted cautiously.

6. Inference Pipeline

The `inference.py` script performs:

1. Load saved model and scaler (`.pkl`)
 2. Load unseen data (`btc_inference_data.csv / eth_inference_data.csv`)
 3. Apply **same preprocessing and feature engineering** as training
 4. Apply **log transformation and scaling**
 5. Generate **predicted next-day close prices**
 6. Compute metrics (MAE, MSE, RMSE, R²) if actual prices are available
 7. Save predictions to CSV
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7. Conclusion

1. **Historical training data** allowed models to perform well (high R^2 , low RMSE).
 2. **Inference on future data** highlights the difficulty of predicting crypto prices due to:
 - High volatility
 - Non-linear and unpredictable market behavior
 3. BTC predictions are **less reliable**, ETH predictions are **moderately reliable**.
 4. Crypto price prediction should be used for **trend insights**, not exact forecasting.
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8. Future Work

- Include **additional market indicators** (e.g., sentiment analysis, macroeconomic data).
- Explore **deep learning models** like LSTM or Transformer-based architectures.
- Implement **dynamic retraining** to adapt models to evolving crypto patterns.
- Quantify **prediction confidence intervals** to better understand uncertainty.